

Math 1232 Summer 2025

Mastery Quiz 2

Due Tuesday, July 8

This mastery quiz has three topics. Everyone should do M1 and S2, but if you already have a 2/2 on S1, feel free to skip it. Please do your best, but don't worry if you make a minor error—your goal is to demonstrate your mastery of the underlying material.

Feel free to consult your notes, but please don't discuss the actual quiz questions with other students in the course.

Remember that you are trying to demonstrate that you understand the concepts involved. For all these problems, justify your answers and explain how you reached them. Do not just write “yes” or “no” or give a single number.

Please turn this quiz in class on Tuesday. You may print this document out and write on it, or you may submit your work on separate paper; in either case make sure your name is clearly on it. If you absolutely cannot turn it in in person, you can submit it electronically but this should be a last resort.

Topics on this quiz:

- Major Topic 1: Transcendental Functions
- Secondary Topic 1: Invertible Functions
- Secondary Topic 2: L'Hôpital's Rule

Name: Solutions

Major Topic 1: Transcendental Functions

(a) Compute $\frac{d}{dx}x^{\ln(x)}$.

Solution. The simplest approach is to use logarithmic differentiation.

$$\begin{aligned}y &= x^{\ln(x)} \\ \ln(y) &= \ln(x) \ln(x) = \ln(x)^2 \\ \frac{y'}{y} &= 2 \ln(x) \frac{1}{x} \\ y' &= \frac{2 \ln(x)}{x} y = \frac{2 \ln(x) x^{\ln(x)}}{x}.\end{aligned}$$

(b) Compute $\frac{d}{dx} \frac{1}{\arcsin(x^2)}$.

Solution.

$$\frac{d}{dx} \frac{1}{\arcsin(x^2)} = \frac{-1}{(\arcsin(x^2))^2} \cdot \frac{1}{\sqrt{1-(x^2)^2}} \cdot 2x.$$

(c) Compute $\int_0^2 \frac{e^{2x}}{e^{4x}+1} dx$.

Solution. We can take $u = e^{2x}$ so $du = 2e^{2x} dx$ and

$$\begin{aligned}\int_0^2 \frac{e^{2x}}{e^{4x}+1} dx &= \int_1^{e^4} \frac{1}{2} \frac{1}{u^2+1} du = \frac{1}{2} \arctan(u) \Big|_1^{e^4} \\ &= \frac{1}{2} \arctan(e^4) - \frac{1}{2} \arctan(1) = \frac{1}{2} \arctan(e^4) - \frac{\pi}{8}.\end{aligned}$$

Recall that u -substitution with definite integrals is done by changing the 0 and 2 bounds to $u(0) = 1$ and $u(2) = e^4$ bounds, or by finding an antiderivative

$$\int \frac{e^{2x}}{e^{4x}+1} dx = \frac{1}{2} \arctan(e^{2x}) + C$$

and then evaluating at the original bounds to produce the same answer. Note that

$$\int_0^2 \frac{e^{2x}}{e^{4x}+1} dx = \int_0^2 \frac{1}{2} \frac{1}{u^2+1} du = \frac{1}{2} \arctan(u) \Big|_0^2 = \frac{1}{2} \arctan(e^{2x}) \Big|_0^2$$

is *incorrect*, because the first and third equalities are not true—the bounds must be changed.

Secondary Topic 1: Invertible Functions

(a) Find a formula for the inverse of $g(x) = (x - 1)^3 + 3$.

Solution.

$$\begin{aligned}y &= (x - 1)^3 + 3 \\y - 3 &= (x - 1)^3 \\\sqrt[3]{y - 3} &= x - 1 \\x &= 1 + \sqrt[3]{y - 3}\end{aligned}$$

so $g^{-1}(y) = 1 + \sqrt[3]{y - 3}$ or $g^{-1}(x) = 1 + \sqrt[3]{x - 3}$. (You can use whichever variable you like in your formula.)

(b) Compute $\log_3(90) + \log_3(3/2) - \log_3(5)$

Solution.

$$\log_3(90) + \log_3(3/2) - \log_3(5) = \log_3\left(\frac{90 \cdot \frac{3}{2}}{5}\right) = \log_3(27) = 3.$$

(c) Let $h(x) = x^7 + 3x^3 + 1$. Compute $(h^{-1})'(5)$.

Solution. By the Inverse Function Theorem, we know that

$$(h^{-1})'(5) = \frac{1}{h'(h^{-1}(5))}.$$

Guess and check shows that $h(1) = 5$ so $h^{-1}(5) = 1$. And we know that

$$h'(x) = 7x^6 + 9x^2$$

and thus

$$h'(1) = 7 + 9 = 16$$

Thus

$$(h^{-1})'(5) = \frac{1}{16}.$$

Secondary Topic 2: L'Hôpital's Rule

(a) Compute $\lim_{x \rightarrow 2} \frac{e^{(x^2-4)} - x + 1}{x - 2}$.

Solution. Since the numerator and denominator both approach 0, we can use L'Hôpital's Rule, which gives

$$\lim_{x \rightarrow 2} \frac{e^{(x^2-4)} - x + 1}{x - 2} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 2} \frac{e^{(x^2-4)} \cdot 2x - 1}{1} = e^0 \cdot 2 \cdot 2 - 1 = 3.$$

(b) Compute $\lim_{x \rightarrow 0} \frac{x^3 - x^2}{x + \sin(x)}$.

Solution. Since the numerator and denominator both approach zero, we can use L'Hôpital's Rule, which gives

$$\lim_{x \rightarrow 0} \frac{x^3 - x^2}{x + \sin(x)} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow 0} \frac{3x^2 - 2x}{1 + \cos(x)} = \frac{0}{2} = 0.$$

Note that we *cannot* use L'Hôpital's Rule a second time, since the denominator does not approach zero. If we use it, we get the wrong answer that the function approaches $\pm\infty$ and therefore doesn't exist.

(c) Compute $\lim_{x \rightarrow \infty} x^{\frac{3}{2+\ln(x)}}$.

Solution. We use logarithms to turn the exponent indeterminate form ∞^0 into an $\frac{\infty}{\infty}$ indeterminate form:

$$\begin{aligned} \ln y &= \frac{3}{2 + \ln(x)} \ln(x) \\ \lim_{x \rightarrow \infty} \ln y &= \lim_{x \rightarrow \infty} \frac{3 \ln(x)}{2 + \ln(x)} \stackrel{\text{L'H}}{=} \lim_{x \rightarrow \infty} \frac{3/x}{1/x} = 3 \end{aligned}$$

and thus

$$\lim_{x \rightarrow \infty} y = e^3.$$